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Description

This week, we explored regularisation techniques in machine learning, focusing on Ridge (L2) and Lasso (L1) regularisation. I learned how these methods reduce overfitting by adding penalties to model coefficients. We also conducted a mock presentation to practise communicating our data science projects and explaining technical results clearly.

Feelings

At first, I found the difference between Ridge and Lasso slightly confusing because both methods reduce coefficient sizes. After applying them in Python, I understood their differences more clearly. I also felt nervous during the mock presentation because I was concerned about explaining technical concepts in a simple and understandable way.

Evaluation

The activities helped me develop both technical and communication skills. The regularisation exercise allowed me to connect theoretical concepts with practical machine learning. However, I found it difficult to explain technical concepts in a simple way during the mock presentation. This showed me that completing an analysis does not automatically mean I can communicate its results effectively.

Analysis

I realised that selecting a machine learning method requires understanding the characteristics and purpose of the data rather than simply applying an algorithm. Ridge is useful when many features contribute to the model, while Lasso can be useful when feature selection is important. This helped me understand that the choice of regularisation should depend on the modelling goal. The presentation also showed me that effective communication requires a clear structure and explanations suited to the audience.

Conclusion

Overall, I developed a better understanding of how to choose regularisation techniques based on the modelling goal, rather than simply knowing how they work. I also became more aware that technical knowledge alone is not enough to communicate an analysis effectively, making communication an important part of my development as a data science student.

Action Plan

In future, I will practise explaining one machine learning concept in simple language each week and record myself to identify unclear explanations. I will also rehearse presentations before assessments and continue using Python exercises to practise applying and comparing different machine learning techniques, strengthening my technical skills.

